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Beyond continuationism: climate change, economic growth, and the future of world (dis)order

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Abstract *An apocalyptic zeitgeist infuses global life, yet this is only minimally reflected in International Relations (IR) debates about the future of world order and implications of climate change. Instead, most approaches within these literatures follow what I call a “continuationist” bias, which assumes that past trends of economic growth and inter-capitalist competition will continue indefinitely into the future. I identify three key reasons for this assumption: 1) a lack of engagement with evidence that meeting the Paris Agreement targets is incompatible with continuous economic growth; 2) an underestimation of the possibility that failure to meet these targets will unleash irreversible tipping points in the earth system, and 3) limited consideration of the ways climate change will converge with economic stagnation, financial instability, and food system vulnerabilities to intensify systemic risks to the global economy in the near-term and especially later this century. I argue that IR scholars should therefore explore the potential for ‘post-growth’ world orders to stabilize the climate system, consider how world order may adapt to a three or four degree world if Paris Agreement targets are exceeded, and investigate the possible dynamics of global ‘collapse’ in case runaway climate change overwhelms collective adaptation capacities during this century.*

Introduction

An apocalyptic zeitgeist infuses global life, with the spectres of runaway climate change, widespread challenges to multilateral cooperation, and the COVID-19 pandemic feeding a growing sense of a looming era of global turbulence. In 2019 over 11,260 scientists issued a warning that ‘planet earth is facing a climate emergency’ that is ‘accelerating faster than most scientists expected (Ripple et al 2019, 1–2). The Intergovernmental Panel on Climate Change (IPCC) released a report in 2018 that calls for ‘rapid and far-reaching’ changes to meet the Paris Agreement targets of 1.5 or 2 °C above pre-industrial levels (IPCC 2018, 21), considered necessary by some to ward off irreversible tipping points in the earth system (Steffen et al 2018; Lenton et al 2019). Yet a 2019 UNEP report shows that ‘countries’ planned fossil fuel production not only exceeds 1.5 °C and 2 °C pathways’, but also exceeds levels consistent with the implementation of their (already inadequate) pledges under the Paris Agreement (UNEP 2019, 4). On top of this, the latest global climate models suggest that the ‘climate equilibrium sensitivity’ (the amount of warming that would follow a doubling of pre-industrial CO₂ levels) may be higher than previous estimates (Lenton et al 2019, 595). Overall this means that the earth is likely on pace for around four and possibly more degrees of warming by the end of the twenty-first century (Vince 2019),

which Kevin Anderson argues would be ‘incompatible with an organized global community ... likely to be beyond “adaptation” ... devastating to the majority of ecosystems, and has a high probability of not being stable’ (Anderson, quoted in Spratt and Dunlop 2018, 14).

The discipline of International Relations (IR) has not been immune to the apocalyptic zeitgeist, most evident these days in the discourse proclaiming a ‘crisis in the liberal international order’ and others investigating the implications of climate change for world politics. Yet we should ask whether these discourses have an adequate understanding of the challenges posed by the climate crisis and the possible transformations ahead. On one hand, debates about the present crisis and future of world order continue to either ignore climate change or treat it as an issue that can simply be solved by greater global cooperation (e.g. Ikenberry 2018; Haas 2017; Acharya 2015; Buzan and Lawson 2014; Hurrell 2018). On the other hand, while many IR scholars have made climate change central to their analysis of world politics (e.g. Victor 2011; Biermann et al 2012; Galaz 2014; Newell and Paterson 2010; O’Brien et al 2013; Burke et al 2016; Cudworth and Hobden 2011), this literature tends to underestimate its transformative implications for at least three key reasons: (1) a lack of engagement with evidence that meeting the Paris Agreement targets is incompatible with continuous economic growth; (2) an underestimation of the possibility that failure to meet these targets will unleash irreversible tipping points in the earth system, and 3) limited consideration of the ways climate change will converge with economic stagnation, financial instability, and food system vulnerabilities to intensify systemic risks to the global economy in the near-term and especially later this century.

In contrast to these approaches, I will argue that IR scholars should more deeply investigate the potential for structural discontinuities in world order resulting from climate change and its intersection with political-economic, financial and food crises. In particular, most IR scholars have not yet recognized the unprecedented challenge that the climate crisis poses to the growth-based liberal capitalist world order, instead assuming that it can be solved without any structural political-economic transformations in the order itself (e.g. Ikenberry 2018; Buzan and Lawson 2014; Haas 2017; Victor 2011; Biermann et al 2012; Galaz 2014). I use the term ‘continuationism’ to refer to this assumption: that coming decades will see no dramatic transformations of world order outside the basic parameters of economic growth, inter-capitalist competition and the shifting balance of power that have defined the past two centuries of world politics. While some IR scholars *do* recognize the transformative implications of the climate crisis (e.g. Burke et al 2016; Newell and Lane 2020; Dalby 2015; Friedrichs 2013; Grove 2019; Di Muzio 2015), they do not demonstrate why it will likely force an end to economic growth during this century, or explore the possible contours of a more sustainable ‘post-growth’ world order.¹ In contrast, the key contributions of this article will be

¹ The work of Friedrichs (2013) and Di Muzio (2015) come closest in this regard, though they focus primarily on the role of ‘peak oil’ and energy scarcity in forcing an end to growth. While this scenario should not be discounted, the rise of unconventional oil and the current supply glut suggests that oil depletion is unlikely to be the critical factor that cripples the growth trajectory, though they may play an important role in conjunction with climate and other stressors in the future.

to first demonstrate why climate change will likely bring an end to the growth-based liberal capitalist world order, and then to explore three subsequent sets of questions and possible futures that have to date received minimal attention by IR scholars: 1) How might a post-growth world order, capable of meeting the Paris Agreement targets, be designed, and how might it emerge? 2) If a near-term post-growth transition fails to materialize and the Paris targets are exceeded, then how might states and the global economy as a whole adapt to maintain resilience in a 3 or 4°C world? And 3) If states and the global economy are unable to adapt to 3 or 4°C of warming this century, then what would be the possible dynamics of a global-scale ‘collapse’?

I will begin with an overview of recent work on the future of world order and climate change in IR, highlighting the limitations and blind spots that result in continuationist assumptions. I will then elaborate the key reasons why the climate crisis will likely force a structural discontinuity beyond the growth-based liberal capitalist order. Finally, I will explore three possible futures of world (dis)order that may result from this discontinuity, which will suggest possible pathways forward for the study of IR in an age of climate emergency.

Continuationism in the future of world order and climate IR literature

The argument that the modern world system will eventually face an epochal discontinuity in its trajectory of continuous growth has a long lineage, though it became most prominent in debates from the 1970s centring on the ‘limits to growth’. In particular, the (in)famous Club of Rome report from 1972 concluded that

if the present growth trends in world population, industrialization, pollution, food production, and resource depletion continue unchanged, the limits to growth on this planet will be reached sometime within the next one hundred years (Meadows et al 1972, 23–24).

The combination of techno-modernist faith in capitalist markets and innovation, alongside critiques of the original report based on misrepresentations of its actual claims (e.g. the mistaken perception that the report claimed we would ‘run out’ of resources), have led to frequent dismissals of its continuing relevance in the twenty-first century (Higgs 2016, 258–259). Yet multiple studies demonstrate that the world system remains squarely on the ‘business as usual’ path it anticipated, while the looming climate crisis lends support to its conclusion that limits to growth will most likely be reached during the twenty-first century (as I’ll demonstrate below) (Higgs 2016, 266; Jackson and Webster 2016).

While limits to growth thinking has flourished in recent years within the heterodox fields of steady-state and post-growth economics (e.g. Daly 1996; Jackson 2018; Kallis and Hickel 2020; Parrique et al 2019), it has for the most part failed to penetrate the field of IR. In contrast to these fields, IR scholars have rarely considered the potential for discontinuities in the trajectory of endless growth and often appear beholden (whether implicitly or explicitly) to the neoclassical and techno-modernist faith in markets and innovation to sustain this trajectory.² In this sense, the discipline of IR largely follows what I call

² Exceptions to this tendency include Burke et al. 2016; Dalby 2015; Friedrichs 2013; Falk 2016; Grove 2019; Di Muzio 2015

‘continuationist’ assumptions, or the view that there will be no deep transformations in the political-economic structure of world order in the foreseeable future. Continuationists, in this sense, do not expect much change beyond a decline in cooperation and/or shifts in the balance of power between states, and often assume that twenty-first century problems can be solved without any deep political-economic transformations in world order. Following Richard Falk and Jörg Friedrichs, it characterizes approaches—whether of realist or liberal stripes—that depict the future as ‘essentially confined to an incremental continuation of the past’ (Falk 2016, 102), or that ‘project past economic growth and technological progress’ indefinitely into the future (Friedrichs 2013, vii).

In particular, in this article I focus on approaches that—either explicitly or implicitly—assume that global capitalism and continuous economic growth will remain the taken-for-granted context in which geopolitical competition and/or international institution building takes place. While it has arguably been central to world order since the nineteenth century industrial revolution, the pursuit of economic growth explicitly became a key, arguably *the* key, foundation of liberal world order since it was adopted in the post-World War II era as the primary goal of economic policy and a ‘core state imperative’ (Barry 2020; Pettifor 2019; Schmelzer 2016; Higgs 2016). As Bentley Allan explains, economic growth—understood as annual increases in Gross Domestic Product (GDP)—provided ‘the lynchpin of the design’ of the Bretton Woods system, which enabled the institutionalization of global trade and investment regimes based on the pursuit of collective gains, thereby weakening realist assumptions of a ‘zero-sum’ geopolitical landscape while replacing the ‘explosive ideological conflict between labour and business’ with a ‘broad social consensus on policies to expand production’ (Allan 2018, 222–223). More deeply, economic growth became ‘equated with human progress itself’, involving not simply a key policy goal of states but a worldview that reconceives time as ‘a linear, unending plane upon which economic progress unfolds’ (ibid, 209, 25). While it is rarely identified explicitly as a key component of the contemporary world order, the expectation of continued economic growth constitutes what Donella Meadows calls one of our society’s ‘great but unstated assumptions – unstated because unnecessary to state’ (Meadows, quoted in Pettifor 2019, 72). However, it is precisely this dynamic and assumption of endless growth that is being called into question by contemporary environmental crises, particularly climate change. Yet few scholars of IR, both those investigating the current crisis and future of world order as well as those focused on the implications of climate change for world politics, have begun to seriously consider the possibility that economic growth may come to an end during the twenty-first century.

Starting with the first set of approaches, scholars of ‘world order’ study changes in the practices, institutions, and norms through which particular patterns of inter-state behaviour are regulated and reproduced (Allan 2018, 5; Ikenberry 2011, 13; Haas 2017, 22). While the literature is vast, we can roughly distinguish between approaches that focus on the crisis of the contemporary liberal order and rise of protectionism (Ikenberry 2018; Haas 2017), the shift of geo-economic power to East Asia and potential for counter-hegemonic conflict (Mearsheimer 2018; Layne 2018), and the rise of a more multipolar world order

(Buzan and Lawson 2014; Acharya 2015; Hurrell 2018). John Ikenberry, for one, argues that recent shocks to the liberal order, from Brexit to the Trump election, can be explained as in part a ‘crisis of authority’ and in part a ‘crisis of social purpose’ resulting from the end of the Cold War and the entrance of a wide diversity of states into the order (Ikenberry 2018). Rather than a ‘crisis in the deep principles of the order itself’ (Ikenberry 2011, 6), Ikenberry believes that contemporary problems—from rightwing populism to climate change—are no more challenging than previous threats to the liberal order and can be solved without any transformations in the political-economic principles of the order itself (Ikenberry 2018, 22). Others argue instead that world order is likely to devolve into regional spheres of influence. Buzan and Lawson, for example, argue that we are today witnessing the emergence of a ‘decentred globalism’ in which ‘no state will be able to replace the United States as a superpower, because none will be able to acquire enough relative power to dominate the system as a whole’ (Buzan and Lawson 2014, 75). They claim that this will usher in a multipolar world in which the core problematic becomes one of managing the relations between diverse modes of capitalist governance, with the hegemony of capitalist economic organization being firmly settled for the foreseeable future (ibid, 72). While realists argue that the decline of American leadership will usher in a world of renewed regional rivalry and potentially hegemonic war (Mearsheimer 2018; Layne 2018), others argue that a more multi-polar or ‘posthegemonic’ world offers opportunity for more egalitarian cooperation between the global north and south (Acharya 2015; Hurrell 2018).

The problem with these approaches is not that they are unaware of climate change. Many of them explicitly recognize it as a significant problem, though they tend to believe it will promote cooperative problem-solving and international stability (Ikenberry 2018; Buzan and Lawson 2014; Haas 2017). The primary problem is that they do not consider the possibility that solving the climate crisis may require a radical political-economic restructuring of world order, or that it may force such changes later this century (or sooner) if states fail to reduce emissions quickly enough. Ikenberry, for example, in thinking that contemporary challenges to the order are no greater than past crises, does not grasp the scope or speed of the changes needed to stabilize the climate system. As I will discuss below, evidence suggests this will require more than just rejuvenating international cooperation; rather, a transformation in the ‘deep principles’ of the global economy—particularly the core principle of economic growth—is likely necessary (Kallis and Hickel 2020; Parrique et al 2019; Jackson 2018). Similarly, Buzan and Lawson do not consider the possibility that climate change and its convergence with political-economic crises may threaten assumptions of perpetual economic growth and force more radical political-economic transformations over the course of this century. Even realists who perceive an imminent end to the liberal order do not consider the possibility that capitalism and economic growth may be destabilized and transformed, instead assuming that the current crisis represents merely the continuation of cyclical dynamics of hegemonic rise and fall (Mearsheimer 2018; Layne 2018). Overall, as I will demonstrate below, these scholars have yet to grasp the epochal significance of the climate crisis, at best seeing it as merely one issue among many rather than an existential threat that likely cannot be solved within the framework of a growth-based liberal capitalist world order.

Of course, many IR scholars have made climate change central rather than peripheral (or non-existent) in their analysis of world politics. The emerging climate IR literature can be roughly divided between those focusing on climate governance strategies and mechanisms and those investigating its implications for national and global security. The first set of approaches, as Jennifer Clapp and Peter Dauvergne show, is primarily composed of descriptive approaches illuminating the complex array of international treaties, institutions, and market mechanisms that form a nascent global architecture of climate governance (Clapp and Dauvergne 2016). There are also more prescriptive approaches, with some like David Victor arguing for incrementalist approaches based on carbon taxation imposed by a ‘club’ of key states (Victor 2011, 23). More radical approaches, on the other hand, argue for reform and empowerment of the UN system to ‘mainstream’ environmental goals into economic policy (Galaz 2014; Biermann et al 2012) or to more radically shape global corporations and markets through state intervention to accelerate decarbonization efforts (Newell and Paterson 2010; Dauvergne 2018). The climate security literature is also distinguished by more descriptive and prescriptive approaches. Many of these scholars detail the ways in which national security agencies are preparing for a climatically changed world (Busby 2008). Others investigate the connection between climate impacts and violent conflict in order to assess whether or not climate change will intensify conflict in the future (Mazo 2010; Selby 2014). Finally, the climate security literature also encompasses more critically oriented scholars who focus on the implications of climate change for human security across the domains of agriculture, natural hazards, health and migration, and emphasize the need for more transformative approaches to solve climate change in a truly just and sustainable manner (O’Brien et al 2013; Dalby 2015; Barnett et al 2013; Cudworth and Hobden 2011).

While climate IR scholars often have a deeper appreciation of the challenges confronting world order in the twenty-first century, they also tend to follow continuationist assumptions and have given very limited attention to the possibility that climate change will bring an end to the growth-based liberal capitalist world order. David Victor, for example, proposes minimally disruptive approaches for decarbonizing the global economy, believing that a low and rising carbon tax would be sufficient to manage climate change (Victor 2011, 28). However, as I will elaborate below, evidence demonstrates that such incrementalist reforms would be incapable of reducing emissions with the needed speed to meet the Paris Agreement targets (Anderson and Jewell 2019; Kallis and Hickel 2020). Even more radical approaches—e.g. those of Frank Biermann, Victor Galaz, Matthew Paterson, Peter Newell and Peter Dauvergne—do not systematically consider the possibility that the persistence of compound economic growth may be incompatible with climate stabilization, nor do they investigate how failure to meet the Paris targets may provoke catastrophic transformations in world order.

In the climate security literature, most approaches focus on current and possible future impacts though without foregrounding the growth imperative as a key structural impediment to mitigating the crisis. For example, Karen O’Brien and colleagues claim that it is necessary to identify the ‘root causes’ of the climate crisis (O’Brien et al 2013, 2). Yet they do not identify the growth-based global economy as a key structural cause. Similarly, Jon Barnett and

colleagues, in their anticipations of possible adaptation strategies in a four degree world, do not consider the need to restrain economic growth, and even their best case scenario of radical and equitable adaptation assumes that this will involve ‘dematerialized and decarbonized’ growth (Barnett et al 2013, 229). Simon Dalby and co-authors of the ‘Planet Politics’ manifesto, along with the work of Stephen Hobden and Erika Cudworth, may go furthest by foregrounding the political-economic structures fuelling the climate crisis and the need for rapid transformative action (Dalby 2015; Burke et al 2016; Cudworth and Hobden 2011). However, these scholars do not systematically demonstrate why preventing catastrophic climate change most likely requires an end to economic growth, envision or advocate for more egalitarian ‘post-growth’ world orders that could potentially meet the Paris targets, or think through the likely consequences for world order if a post-growth transition fails to materialize.

Overall, we can identify three key limitations of both future of world order and climate IR scholars: first, that they have yet to systematically engage the evidence that meeting the Paris Agreement targets will require structural political-economic transformations that are incompatible with continuous economic growth; second, that many (though not all) of them ignore or downplay the evidence that failing to meet these targets would unleash irreversible tipping points in the earth system that would be catastrophic for human security and international stability; and third, that they ignore how climate change will feedback on and exacerbate other global systemic risks—e.g. in global economic, financial and food systems—that may trigger global-scale ‘synchronous failures’ in the near-term and especially later this century (Homer-Dixon et al 2015). Once we take this evidence and these feedbacks into account, the challenges facing the liberal-capitalist world order become more daunting, and the need to at least consider the potential for structural political-economic discontinuities—including the possibility of ‘collapse’—becomes evident.

I will now elaborate these three arguments and will then explore their possible implications for the future of world order.

The (likely) impossibility of meeting paris agreement targets in a context of compound economic growth

The assumption that climate change can be solved within a growth-based capitalist framework rests on the hope of ‘decoupling’ economic growth from greenhouse gas emissions by substituting fossil fuels with carbon-free energy and improving energy efficiency. Proponents believe not only that climate protection need not come at the expense of economic growth, but also that it will be a boon to growth through creating jobs and investment opportunities in the energy and ‘green’ industrial sectors (Figueres 2017). However, the problem with these claims is that they do not rigorously consider the question of *speed*, or of whether decoupling from carbon emissions can happen fast enough in a context of compound growth to meet the Paris Agreement targets (Kallis and Hickel 2020; Antal and Van den Bergh 2016; Jackson 2018).

The IPCC claims that emissions need to peak in 2020 and thereafter fall at a rate of 7-8 percent annually, reaching net zero by 2050, to provide a 50 percent chance of meeting the 1.5 °C target, whereas they need to fall 3-4 percent annually, reaching net zero by 2075, to meet the 2 °C target (IPCC 2018, 15).

However, there is a bulk of modelling evidence suggesting that a 7 percent emissions reductions rate would be incompatible with compound economic growth. For example, the C-ROADS model, developed at MIT, projects that the fastest possible rate of decarbonization in a context of compound growth is likely around 4 percent per year, which would require ‘the most aggressive possible abatement policies, high subsidies for renewables and nuclear power, plus high taxes on oil, gas and coal’ (Hickel 2019, 55). Other approaches, including Nicholas Stern’s influential Review of the Economics of Climate Change, similarly conclude that 3-4 percent annual reductions would be the fastest rates compatible with economic growth (Anderson and Bows 2011, 40). While this would be far from capable of meeting the 1.5°C target, such rates may be capable of meeting the 2°C target according to IPCC projections, assuming that an ambitious globally coordinated mitigation program begins in the year 2020 (or very shortly thereafter). However, given the likely conservatism of IPCC projections, which do not consider positive carbon-cycle feedbacks (Steffen et al 2018), there is good reason to be sceptical that 3-4 percent annual decarbonization would be sufficient to meet even the 2°C target. Other approaches accounting for these feedbacks argue that 7 percent reductions per year, reaching net zero emissions by 2050, would give us a 66 percent chance of stabilizing temperature increases at 2°C (Rockstrom et al 2017).

In short, the problem is that so long as the economy continues to grow, the mountain of energy consumption that carbon-free energy must conquer only gets higher, requiring even faster rates of decarbonization. Kevin Anderson starkly summarizes the implications of these findings:

Ongoing failure to mitigate emissions has pushed the challenge from a moderate change in the economic system to a *revolutionary overhaul of the system*. This is not an ideological position; it emerges directly from a scientific and mathematical interpretation of the Paris climate agreement (Anderson and Jewell 2019, 348; italics added).

Technological optimists would counter that climate stabilization can be made compatible with economic growth by ramping up ‘Negative Emissions Technologies’ (NETs), including direct air capture, Bioenergy plus Carbon Capture and Sequestration (BECCS), afforestation and reforestation. Indeed, all of the IPCC’s ‘Representative Concentration Pathways’ compatible with 1.5 or 2°C rely on a significant rollout of NETs (IPCC 2018). However, the problem is not only that some of these technologies are presently hypothetical and unproven at scale, but also that they would entail massive land, energy and water requirements to put a dent in global emissions (i.e. removing five gigatons or more of carbon per year by 2050 and up to twenty annually by 2100). For example, the IPCC’s fifth assessment report estimates that the scale of BECCS plants needed would ‘require plantations covering land two to three times the size of India, which raises questions about land availability, competition with food production, carbon neutrality and biodiversity loss’ (Hickel and Kallis 2020, 10). Furthermore, the costs and energy requirements for direct air capture are ‘currently prohibitive and can be anticipated to slow deployment’, which would make this technology very difficult to scale up rapidly enough to meet the Paris targets (Smith et al 2016, 48). Such schemes might be possible through a massive globally coordinated effort, and technological innovation is

to be expected. But those who confidently expect NETs to eventually make climate stabilization compatible with economic growth, simply put, are ‘taking a very large risk with our common future’ (Antal and Van den Bergh 2016, 171).

The potential for irreversible tipping points if we cross 1.5 or 2 °C

Due to the challenges of rapidly reducing carbon emissions in a context of economic growth, economists and climate policy analysts are increasingly arguing that we need to prepare for a hotter-than-2°C world (e.g. Nordhaus 2013). From a *gradualist* or linear understanding of climate change, such an approach may appear reasonable, even though 3 °C is projected to double the number of people exposed to overlapping climate stresses relative to a 2 °C world, with the vast majority of them concentrated in the global south (Byers et al 2018). However, the potential for positive feedbacks unleashing a trajectory of runaway warming, which numerous planetary scientists believe will become more likely as we cross 1.5 and 2 °C, reinforces the importance of meeting the Paris Agreement targets (Lenton et al 2019; Steffen et al 2018; Wadhams 2017).

Due to scientific uncertainty and lack of precision about the dynamics of tipping elements in the earth system—including ice albedo reduction, tropical and boreal forest dieback, and permafrost carbon and methane release—these dynamics have been excluded in IPCC projections of future warming (IPCC 2018, 103-104). Yet Tim Lenton and colleagues show there is mounting evidence that many of these tipping elements are already being activated—seen in accelerating arctic ice loss and some tropical forests becoming net sources rather than sinks of CO₂—and will be increasingly active at 1.5 and 2 °C (Lenton et al 2019). Their analysis supports the conclusion of Will Steffen and colleagues who consider 2 °C as the point beyond which the earth system would enter a ‘Hothouse Earth pathway’, defined as a state in which

biogeophysical feedbacks in the Earth System ... become the dominant processes controlling the system’s trajectory ... raising the temperature further to activate other tipping elements in a domino-like cascade that could take the Earth System to even higher temperatures (Steffen et al 2018, 3).

The potential for unleashing a hothouse earth pathway is further supported by evidence that the risk posed by carbon and methane release from both sub-sea and terrestrial permafrost may be under-estimated by mainstream climate science (Wadhams 2017; Farquharson et al 2019; Neumann et al 2019). For example, Peter Wadhams argues that the loss of arctic sea ice will likely accelerate the melting of sub-sea methane hydrates by removing the ‘air conditioning layer’ that previously kept arctic waters close to 0 °C (Wadhams 2017, 123). This could unleash an abrupt methane ‘pulse’ of fifty gigatons from the shallow East Siberian Shelf that rapidly raises global temperatures by 0.6 °C, which would ‘speed up all the other global warming effects’, ‘bring forward by fifteen to thirty-five years’ the date at which the global temperature increase exceeds 2 °C (getting us there between 2035 and 2040), and potentially unleash a chain reaction of further hydrate collapse and methane release (ibid, 125-126). Other scientists believe that a catastrophic methane release from sub-sea hydrates is unlikely due to microbial consumption and other ‘sinks’ that

prevent sub-sea methane from reaching the atmosphere (Ruppel and Kessler 2017). However, evidence suggests that the microbial sink may be less effective than previously thought (Thurber et al 2020), and it remains uncertain to what extent these sinks could counteract widespread hydrate collapse and methane release in a $2^{\circ}\text{C}+$ world. The terrestrial permafrost may pose a more long-term threat, though recent studies show that it is already thawing much faster than IPCC models anticipate due to ‘abrupt thawing’ events associated with thermokarst lakes (Farquharson et al 2019), and will likely continue to thaw more quickly than expected due to increased precipitation over the arctic, driven by higher temperatures (Neumann et al 2019).

In sum, cutting edge research on permafrost, methane hydrates, and other tipping elements continues to demonstrate the likely conservatism of IPCC projections. This is not surprising given that, as Naomi Oreskes and colleagues observe, IPCC projections ‘have systematically underestimated key climate change drivers and impacts’, which they attribute to the tendency among scientists of ‘erring on the side of less rather than more alarming predictions’ (Oreskes et al 2013, 327-328). However, due to the limits of scientific knowledge about these processes, it is impossible to say with any certainty what the risks are. We should not conclude that the worst-case ‘methane bomb’ scenarios are the most likely to materialize (though neither should they be discounted) (Wadhams 2017), but evidence indicates that exceeding 2°C of warming would likely trigger significant carbon and methane feedbacks over the course of this century and in subsequent centuries that would make it difficult if not impossible to stabilize the climate system (even with the aid of NETs).

The convergence of climate change with economic, financial, and food crises

The threat of climate change *by itself* may not pose catastrophic threats to currently powerful states until later this century (though it is already destroying livelihoods and adding further stress to conflict zones around the world, especially in the global south) (Ahmed 2017). However, once we consider how these impacts will feedback on existing vulnerabilities in global economic, financial and food systems, the likelihood of *near-term* systemic crises increases. As Will Steffen and Aled Jones warn, the global political-economy

has been designed around assumptions of a stable world ... But it now sits on top of a hugely unstable and complex platform – our physical world, increasingly disrupted by climate change ... we know relatively little about just how fragile the various parts of our clockwork global economy are in the near-term (Steffen and Jones 2019).

Steffen and Jones build upon work done by Thomas Homer-Dixon and colleagues on what they call the emerging ‘architecture of global crisis’, which refers to an intricate structure of causal feedbacks between financial, food and energy systems in which shocks in one sub-system can propagate outwards into ‘synchronous failure’ for the global economy as a whole (as in the 2007-2008 financial crisis) (Homer-Dixon et al 2015). Their analysis can help us anticipate the possible dynamics and trajectories of future crises, but we must

go further by anticipating how this crisis architecture will intersect with climate impacts over the coming years and decades.

First, many analysts show that the global economy is in a structurally precarious position where mediocre growth is reliant on unprecedented debt expansion and a corresponding build-up of financial systemic risk, which may be further exacerbated by the COVID-19 pandemic and the crisis-fighting measures adopted by states and central banks (Ahmed 2017; Mason 2020). Once we bring climate change into the equation, the risks of prolonged stagnation and financial instability intensify. The economic impacts of climate change include capital shifted from production to infrastructure repair and adaptation, heightened uncertainty for investors and consumers, increasing health care costs and diminished labour productivity (particularly for outdoor workers) (Burke et al 2018). There is great uncertainty concerning projected total damages in different mitigation scenarios, though many believe they are currently being under-estimated in most Integrated Assessment Models (e.g. Nordhaus 2013), due mainly to their inability to represent potential tipping points in earth and human systems, ignorance of how climate impacts will constrain economic growth, and assumption that losses from climate impacts are perfectly substitutable with increased consumption and can be fully compensated by higher incomes (Diaz and Moore 2017). Taking into account the feedbacks on economic growth, Marshall Burke and colleagues estimate that a 2.5-3°C temperature rise by 2100 would reduce per capita output by 15-25 percent relative to a world without warming, and by more than 30 percent with 4°C warming, though even these projections do not account for tipping points (Burke et al 2018). Extreme natural disasters will also pose risks to financial stability that are poorly understood. The Cambridge Centre for Risk Studies anticipates that superstorms in New York and Florida could each produce over \$1 trillion in losses from infrastructure damages and their spill-over effects on real estate values, neighbouring regions and trading partners, which could generate further financial instability by triggering a higher-than-expected frequency of insurance claims and debt defaults among affected homeowners and businesses (Mahalingamet al 2018). They note that these events would be unlikely to trigger a financial crisis and/or global recession on their own, though they also assume that markets would 'react rationally and proportionately' (ibid, 8), which suggests that, especially in an environment of high debt and weak growth, more destabilizing dynamics should not be discounted. Additionally, rising temperatures will increase the probability of multiple extreme events occurring in a short time frame, as well as the probability that such events would coincide with ongoing recessions or financial crises, in which case 'the consequences could be very much more severe' (ibid: 7-8).

Second, the global food system is already under threat from unsustainable groundwater depletion, soil degradation and the increasing homogeneity of crop varieties and production techniques designed for efficiency rather than resilience, which will be further stressed by the climate crisis (Cribb 2019; Ahmed 2019). The IPCC projects that, for each degree Celsius of temperature increase, wheat yields will be reduced between 2.9 and 6 percent, rice by 3.2-3.7 percent, maize by 4.5-7.4 percent, and soybeans by 3.1 percent (IPCC 2018, 236). Intensified drought and unsustainable groundwater depletion will also have critical implications that are not factored into the IPCC's projections. For

example, a recent study projects that 40 percent of all irrigated crops will face ‘acute water stress’ as early as 2040 (Ahmed 2019). Around the same time, the National Center for Atmospheric Research projects that nearly all of the US and much of Europe could regularly face ‘extreme drought’ conditions, and large parts of Asia responsible for producing much of the world’s wheat and maize may face similar conditions even earlier (ibid). The implications for overall crop yields are uncertain, though they signify that the IPCC’s projections—which also exclude the impacts of extreme weather, pests and declining insect pollinators—are likely to be conservative (IPCC 2018, 237). Overall, food security analysts believe that crop yields will need to double between 2005 and 2060 to meet increasing demand from population growth (Cribb 2019, 64). However, Challinor et al analysing data from over 1,700 models, demonstrate ‘a majority consensus that yield changes will be negative from the 2030s onwards’ (Challinor et al 2014, 289), suggesting that meeting growing demand will be a huge political-economic and technical challenge, and that global food system shocks may happen sooner than many analysts believe (Ahmed 2019).

Overall, due to existing (and mounting) vulnerabilities in financial and food systems, climate impacts already pose the threat of cascading global crises, which will only intensify at 1.5-2 (and especially 3-4) °C. As Homer-Dixon and colleagues show, any crisis in one sub-system would likely cascade outwards into shocks for the other sub-systems (Homer-Dixon et al 2015). For example, a financial crisis could spark food crises by fuelling speculation on land and agricultural commodities (deemed ‘safe assets’ in times of uncertainty), thereby further exacerbating price rises (as we saw in 2007-2008) (ibid). Conversely, food system shocks could spark a financial crisis and recession by triggering widespread social unrest, weakening consumer demand and undermining investor confidence. While such crises could spur the kind of emergency mobilization needed to accelerate emissions reductions, they may also reinforce a ‘growth at all costs’ mindset, thereby further exacerbating climate change. The COVID-19 crisis is currently demonstrating how vulnerable the global economy is to cascading crises across ecological, financial, food and energy systems, and we can expect that similar shocks will become more likely and severe if global temperatures are not stabilized at 1.5 or 2 °C.

Implications for the future of world (dis)order

Given the likely impossibility of meeting the Paris Agreement targets in a context of compound growth, the likelihood of near-term climate tipping points, and the risks they pose to our already fragile political-economic, financial and food systems, what then are the likely implications for the future of world order? I will suggest that these conclusions should lead us to investigate three possible futures and associated questions that have only received minimal attention by IR scholars: 1) Is a near-term transition to a “post-growth” world order possible to stabilize the climate system? If so, how might it be designed and implemented? 2) If we accept that humanity is on course for three or four degrees of warming this century, then how might states and the global economy as a whole adapt to maintain resilience and prevent mass mortality? And 3) If states and the global economy are unable to adapt to 3 or 4 °C, then what would be the possible dynamics of a global-scale ‘collapse’?

First, if it is true that a transition to a post-growth world order is necessary to meet the Paris Agreement targets, then it is important to consider what this could look like and how it might come about (however unrealistic it may appear). This should lead liberal IR theorists interested in global institutional design, as well as constructivists interested in the influence of ideational change on world order (e.g. Allan 2018), to engage with scholars working in the heterodox fields of steady-state and post-growth economics (Daly 1996; Hickel 2019; Jackson 2018; Pettifor 2019). These approaches share two key arguments: 1) that perpetual economic growth is incompatible with climate stabilization and impossible on a finite planet, since evidence suggests GDP growth cannot be 'absolutely decoupled' from growing resource use and associated environmental impacts (Kallis and Hickel 2020; Parrique et al 2019); and 2) that, beyond a certain threshold, growth ceases to be necessary for improving collective welfare and causes more harm than benefit (Daly 1996; Jackson 2018). A post-growth political-economy, in contrast, would involve global caps on resource consumption, new metrics to measure economic welfare (e.g. the 'Sustainable Development Index' or 'Genuine Progress Indicator'), redistribution through progressive taxation, new public ownership structures that prioritize ends other than shareholder value, a reduction in the work week and greater emphasis on employment in the social and non-market economy, and more radical welfare policies (e.g. 'Universal Basic Income' or the expansion of free public services) to reduce poverty and improve collective welfare in the absence of economic growth (Hickel 2019; Pettifor 2019; Jackson 2018).

Overall, a post-growth transition would constitute a structural political-economic transformation that radically reconfigures the power relations, world-views and goals of states and world order as a whole. As Ann Pettifor describes, it would require an overhaul of Bretton Woods institutions, along with the global assemblage of multilateral and bilateral trade agreements, to ensure equitably shared global consumption caps, reconfigure trade by prioritizing national self-sufficiency in the production of goods where possible, institutionalize capital controls to enhance national autonomy over economic policy and overhaul global finance by consigning the powers of credit-creation to democratically accountable public and central banks (Pettifor 2019, 74-77, 91). This could be considered a form of progressive 'deglobalization' in which supply-chains are relocalized, capital mobility is weakened and economies are re-oriented towards sustainable production for local needs rather than financialization and export-led growth. While radical, elements of this agenda (particularly supply-chain relocalization) are being pursued today in response to the COVID-19 pandemic (Gray 2020), and the growing movement to shrink and democratize the financial sector may find a window of opportunity for transformation in the wake of another (arguably inevitable) financial crisis (Pettifor 2019, 64). Furthermore, the OECD's 'Beyond Growth' initiative and growing scepticism towards GDP among influential economists and heads of state demonstrate that cracks are emerging in the growth consensus that may be creating the preconditions for an eventual post-growth world order (OECD 2019). While there are challenging questions on how a post-growth transition could materialize in a competitive international system in which growing GDP is critical to economic and military power (Barry 2020), it is possible in the coming years that a convergence between political-economic and climate-

induced crises will overwhelm the efforts of leading states to restore economic growth. Combined with strong pressure from grassroots movements like Extinction Rebellion and the Climate Justice Alliance to abandon GDP in favour of a just emergency transition,³ this context may enable the emergence of new international norms and institutions to rapidly reduce emissions and redesign economies around the principles of well-being and steady-state economics.

Second, if we dismiss the feasibility of a near-term post-growth transition to stabilize the climate system, then we should consider the implications of a three or four degree temperature rise for world order and how the latter may adapt to retain stability and prevent mass mortality. While there is much uncertainty, current projections are that agriculture across the planet would be severely threatened in a three and especially four degree world (possibly involving 20-40 percent yield reductions in key staples), which multiple scientists believe could only support a global population of four billion or less (Vince 2019; Cribb 2019, 79). Furthermore, scientists project that, due to a combination of sea level rise and population growth, up to 300 million people will be at risk of displacement from regular coastal flooding by 2050 (Kulp and Strauss 2019), while populations across North Africa, the Middle East and South Asia would be regularly subjected to lethal ‘wet-bulb temperatures’ (the 35°C threshold beyond which sweat-based cooling mechanisms lose their effectiveness even for healthy individuals) (Raymond et al 2020). Therefore, preventing international conflict and mass mortality in this context would likely require ‘cooperating as never before to radically reorganize our world’ through massive programs of resettlement and relocation to shift population centres and agricultural production to more habitable regions (Vince 2019). There is great uncertainty regarding projections of total ‘climate refugees’ in the future, which range widely from 40 million to 1 billion by 2050 (and would likely increase by the time we reach 3-4°C), though much of this would involve domestic rather than international migration (Cribb 2019, 170-171). Entire states would also need to be relocated, particularly the low-lying island nations of the Maldives, Marshall Islands and Kiribati, which would be submerged—along with much of Bangladesh—by the 1-2 metre sea level rise projected by 2100 (Femia and Werrell 2017). Whether and how this could be peacefully managed, particularly in a context of rising rightwing populism and food-water-economic crises stressing all regions of the international system, is a daunting challenge.

However, the most difficult challenge in a three to four degree world may be how to prevent warming from escalating ever further, given the likelihood of tipping elements being intensified across the earth system by this time (Lenton et al 2019). Luck will undoubtedly be necessary, in particular that

³ Climate justice groups like the Climate Justice Alliance have long called for abandoning GDP in favour of alternative notions of well-being, while leading Extinction Rebellion strategists agree that this will be necessary to meet the Paris Agreement targets (Read & Alexander 2020, 16, 23). Furthermore, survey data suggests that the wider public may already be receptive to such demands. A recent YouGov poll, for example, found that 8 in 10 people in the UK would prefer the government to ‘prioritize health and wellbeing over economic growth’ during the COVID-19 pandemic, whereas 6 in 10 would prefer to *permanently* prioritize health and wellbeing over GDP (Harvey 2020).

carbon and methane feedbacks will unfold slowly. This may provide states the breathing room they need to coordinate a global plan for accelerated NET expansion and possibly even temporary Solar Radiation Management (SRM) to hold global temperatures constant and prevent a cascade of tipping points as atmospheric CO₂ levels are brought down. However, as many critics have demonstrated, SRM poses immense logistical and governance challenges—not only due to the geographically uneven distribution of costs and benefits and inherent climatic uncertainties, but also the need to sustain global cooperation to coordinate deployment and maintenance of a geographically dispersed SRM infrastructure for decades or longer to avoid ‘termination shock’ (McKinnon 2020). Even if successful, such adaptations would be unlikely to enable states and the global economy to sustain compound growth. Given the immense economic damages and productivity losses that would occur in a 3–4°C world (Burke et al 2018), the fact that many ecosystems are already on the brink of collapse at current levels of global resource consumption, and evidence that an ‘absolute decoupling’ of economic growth from growing resource consumption is likely impossible, limits to growth will most likely be reached by this time (Kallis and Hickel 2020; Parrique et al 2019). However, a transition to a post-growth global economy, combined with sustained global cooperation to massively expand carbon removal, permanently resettle climate refugees in more habitable regions, relocate and transform global food production and govern a temporary SRM intervention, might enable a stable though radically altered world order to emerge (Vince 2019; Cribb 2019, 259).

Finally, if, as many scientists fear, positive feedbacks in the earth system will raise temperatures more rapidly and overwhelm collective adaptation capacities, then how might a process of global ‘collapse’ unfold? Collapse is understood here, following Joseph Tainter, as a ‘pronounced loss of an established level of sociopolitical complexity’, leading to declines in material standards of living and population levels (Tainter 1988, 193). One key question is whether a global collapse would involve a slow and geographically uneven ‘long descent’ punctuated by a series of severe crises over the course of decades (Greer 2008), or a more rapid process of ‘synchronous failure’ where mutually reinforcing ecological and political-economic crises cripple critical infrastructures and governance capacities (Homer-Dixon et al 2015). This will depend on the precipitating ‘trigger’ and the strength of amplifying feedbacks. For example, a major financial crisis in the next 10–20 years could converge with intensifying climate impacts—whether trillion dollar natural catastrophes or crop failures in key breadbaskets—in which case efforts to stabilize the financial system and restore investor confidence may confront insurmountable headwinds. The result may be a prolonged depression, critical infrastructure failure as governments at national and local scales are unable to mobilize the resources to manage these multiple crises simultaneously, and widespread social unrest (Ahmed 2017).

Such risks would be increasingly pronounced as we reach three and four degrees later this century, especially if realists are correct that global cooperation will continue to fragment in the coming decades (Mearsheimer 2018). In a context of limited international cooperation and non-linear climate change that overwhelms incremental adaptation, this would mean agricultural collapse—potentially ‘reducing world grain production by about one third at the

very time we are trying to double it’—and unprecedented food insecurity, intra-state conflict and geopolitical tension across the international system (Cribb 2019, 78; Spratt and Dunlop 2018). The global economic and financial architecture based on assumptions of continuous growth would likely collapse, precipitating a breakdown of global trade and exacerbating resource scarcities in import-dependent states. Great powers (especially the US and China) may engage in ‘predatory militarism’ to secure food, water and energy supplies for their populations (Friedrichs 2013, 96), and there is a significant danger that inter-state conflict over dwindling resources could escalate into a nuclear exchange. India-Pakistan is a major flashpoint, since both countries are not only highly vulnerable to extreme heat, water stress and crop failures but could also see tension flare up over access to the shared Indus river (Cribb 2019, 66). Thus, we should not downplay the risk that global collapse would lead desperate states to flaunt international norms on conflict prevention and even nuclear deterrence. However, rather than assuming inevitable conflict, it may also generate new forms of cooperation to develop more simplified and localized political-economies based on rationing, redistribution and securing basic needs (Friedrichs 2013, 98).

Conclusion

To conclude, scholars of the current crisis and future of world order should make the climate crisis *central* rather than peripheral (or non-existent) in their analysis due to its likely transformative and/or catastrophic implications. Furthermore, climate IR scholars should more deeply investigate the potential for structural political-economic discontinuities induced by the climate crisis and its feedbacks on political-economic, financial, food and other critical infrastructure systems and explore their geopolitical ramifications.

To clarify, my argument is not that economic growth will *inevitably* come to end during this century. Unanticipated technological breakthroughs in renewable energy, battery storage, next generation nuclear reactors and biofuels *may* enable a more rapid transition to net zero emissions than current models anticipate, thereby making the Paris Agreement targets compatible with economic growth. On the other hand, major breakthroughs in NETs, making them both cost effective and land-energy-water efficient, *may* enable us to overshoot the Paris Agreement targets while bringing atmospheric CO₂ back down to safe levels later this century. But it would be irresponsible to dismiss such scenarios completely due to confident expectations of future technological and political-economic solutions. Instead we need to grapple with the full range of possibilities, especially as climate action continues to be delayed ever further and evidence of near-term tipping points mounts.

The COVID-19 pandemic has demonstrated how systemic shocks made possible by the combination of ecological degradation and political-economic fragilities can rapidly upend global capitalism and world order (Settele et al 2020). While the near-term implications will involve the worst global economic crisis since at least the 1930s great depression, in the mid and longer term the likelihood of either a post-growth transition or global collapse have increased. On one hand, the shock and subsequent lockdowns may permanently alter patterns of production and consumption, thereby changing the values and

expectations of citizens and policy-makers to prioritize well-being, security and ecological protection over rising GDP (Gray 2020). On the other hand, efforts to revive the growth engine at the expense of climate mitigation and other ecological considerations may further entrench a long-term trajectory of ecological and political-economic collapse, especially if the crisis precipitates further concentrations of wealth and a return to financial austerity in the coming years. The current crisis should also make us reflect on how future pandemics—which may be more lethal than COVID-19 (Settele et al 2020)—will converge with intensifying stresses posed by a continuously warming climate on political-economic, financial, food and energy systems. In short, the COVID-19 crisis may provide a glimpse of how synchronous failures emerging from the climate-financial-food-energy-public health nexus may overwhelm the resilience of global capitalism and currently powerful states in the coming years and especially later this century. Therefore, the project of thinking beyond the continuationist view that capitalism and economic growth will persist indefinitely, even as its ecological foundations deteriorate, is more pressing than ever. In contrast, IR scholars should be open to more radically discontinuous trajectories in the international system, and the task of imagining a post-growth world order should be taken up by those hoping to actualize more just and sustainable political-economies from the unravelling of the current order.

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